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Ans 1 Newton's 2nd law: Force applied is equal to rate of change of momentum.

F = dp/dt = m*(v - u)/t = ma.

Therefore F = ma.

Ans 2 m = 1500 kg, u = 0, v = 20 m/s, t = 10 s.

a = (20 - 0)/10 = 2 m/s^2.

Force = mass * acceleration = 1500 * 2 = 3000.

[Note: Student forgot unit 'N']

Ans 3 Work is the energy transferred when a force moves an object.

W = F * s = 50 N * 6 m = 300 J.

Ans 4 Kinetic energy: Energy in motion. E.g. running athlete.

Potential energy: Energy stored due to height. E.g. stretched rubber band.